CSE-251

Electronic Circuits

Lecture-05

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Today's Overview:

- ❑ *Mass Action Law*
- ❑ *Diffusion and Drift of Carriers*
- ❑ *p-n junction DIODE*
- ❑ *Biasing of p-n Junction*

Mass Action Law

Mass Action Law:

At thermal equilibrium and a constant temperature, the product of the number of electrons in the conduction band of a semiconductor material and the number of holes in the valence band will be constant regardless the level of doping concentration.

It is expressed as:

$$\boxed{n_e n_h = \text{constant} = n_i^2 \text{ (function of temperature)}}$$

- n_e= No. of electrons per unit volume in conduction band
- n_h= No. of holes per unit volume in valence band
- n_i= intrinsic concentration per unit volume

Problem:

A Si crystal is doped by arsenic (As) with a doping concentration of 1 ppm. If the crystal contains 5×10^{28} m^{-3} atoms and n_i=1.5×10^{16} m^{-3}, then calculate the number of electrons and holes in the crystal.

Solution:

The doping concentration of semiconductor material is 1 ppm, means 1 part per million (10^6).

That is, 1 atom of As is doped in 10^6 atoms of Si. Therefore, total number of dopant atoms will be,

$$n_{As} = \frac{5 \times 10^{28}\ \mathrm{m}^{-3}}{10^6} = 5 \times 10^{22}\ \mathrm{m}^{-3}$$

We know, one As atom produce one free electron. Thus, the total number of electrons created by doping is,

$$n_e = 5 \times 10^{22}\ \mathrm{m}^{-3}\ [\mathrm{Ans.}]$$

Again we know, $$n_e n_h = n_i^2$$

or, $$n_h = \frac{n_i^2}{n_e} = \frac{(1.5 \times 10^{16}\ \mathrm{m}^{-3})^2}{5 \times 10^{22}\ \mathrm{m}^{-3}} = 4.5 \times 10^9\ \mathrm{m}^{-3}\ [\mathrm{Ans.}]$$

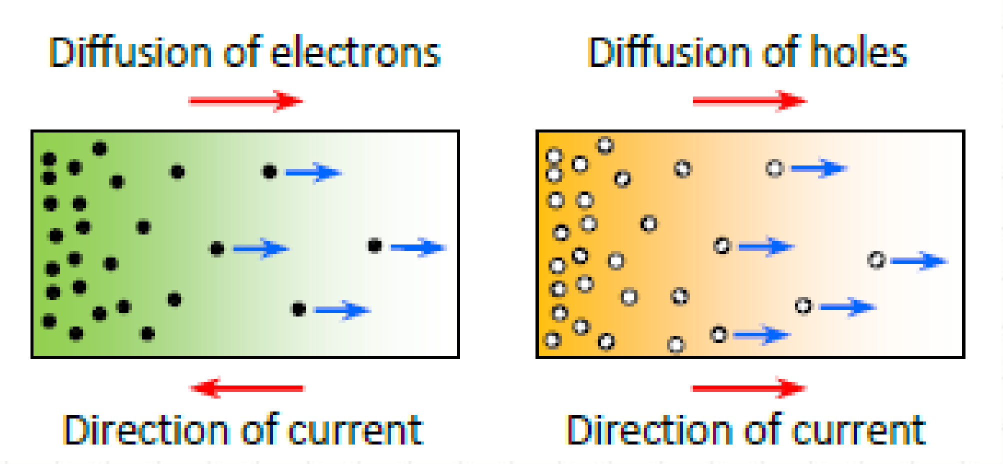
Diffusion and Drift of Carriers

Diffusion:

The movement of charge carriers (in fact any particles) from higher concentration region to lower concentration region is called diffusion.

When electrons or holes move from one place to another by the process of diffusion, electrical current is produced. This current is called ***diffusion current***.

The conventional current direction will be the same as the direction of holes diffusion. On the other hand, the conventional current direction will be opposite to the direction of electrons diffusion.

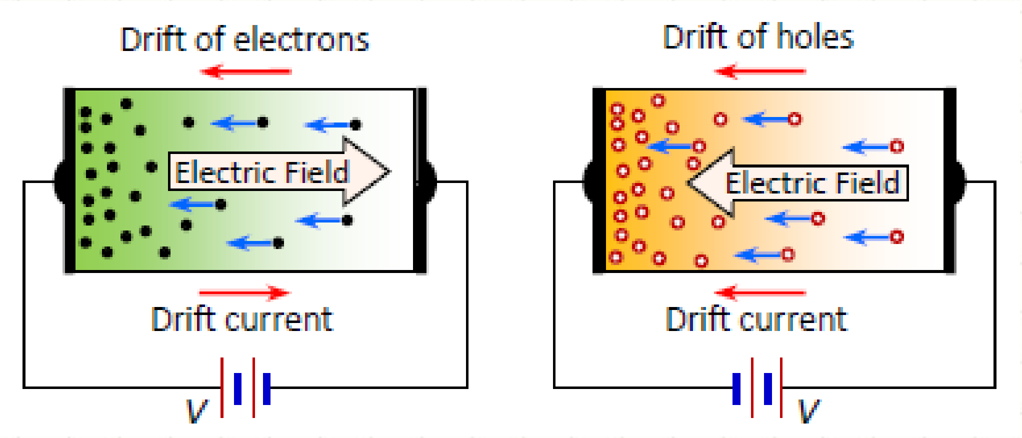


Drift:

The movement of charge carriers (in fact any particles) with the help of any force is called drift.

We know that if a free electron is placed in an electric field, it moves to opposite direction of the electric field due to the force exerted by the eclectic field on the electron. Similarly, if a positive charge (E.g., holes) is placed in an electric field, it moves to the same direction of the electric field due to the force produced by the eclectic field on the holes.

This type of motion of electrons or holes is called drift and the current produced by this process is called drift current.

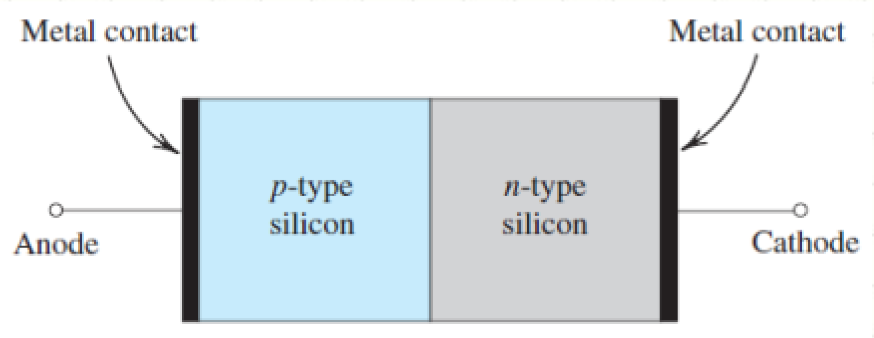


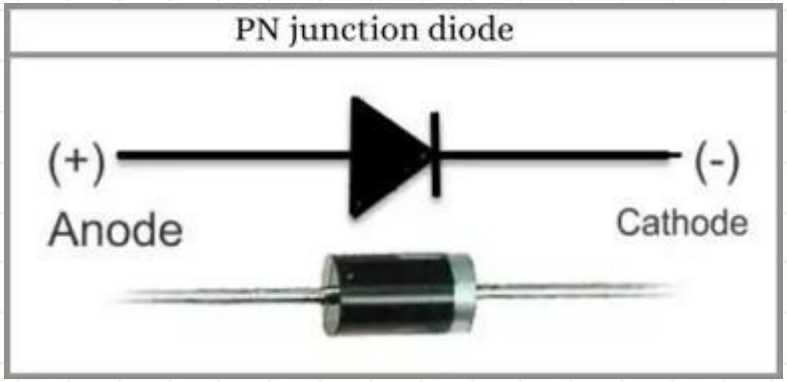
p-n junction DIODE

p-n Junction:

A p-n junction is a boundary between two semiconductor material types, namely the p-type and the n-type, inside a semiconductor.

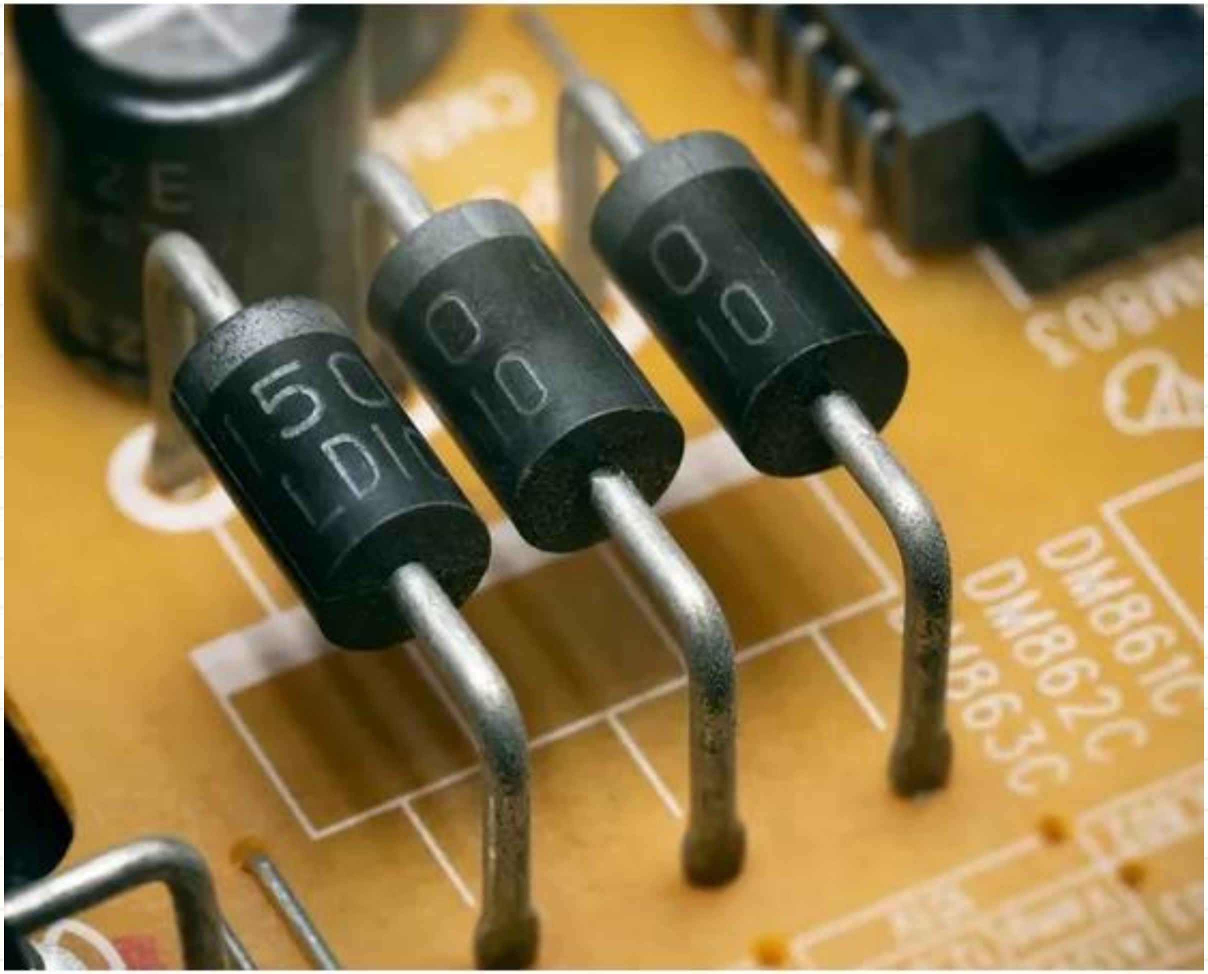
In a semiconductor, the p-n junction is created by the method of doping. The p-side or the positive side of the semiconductor has an excess of holes, and the n-side or the negative side has an excess of electrons.





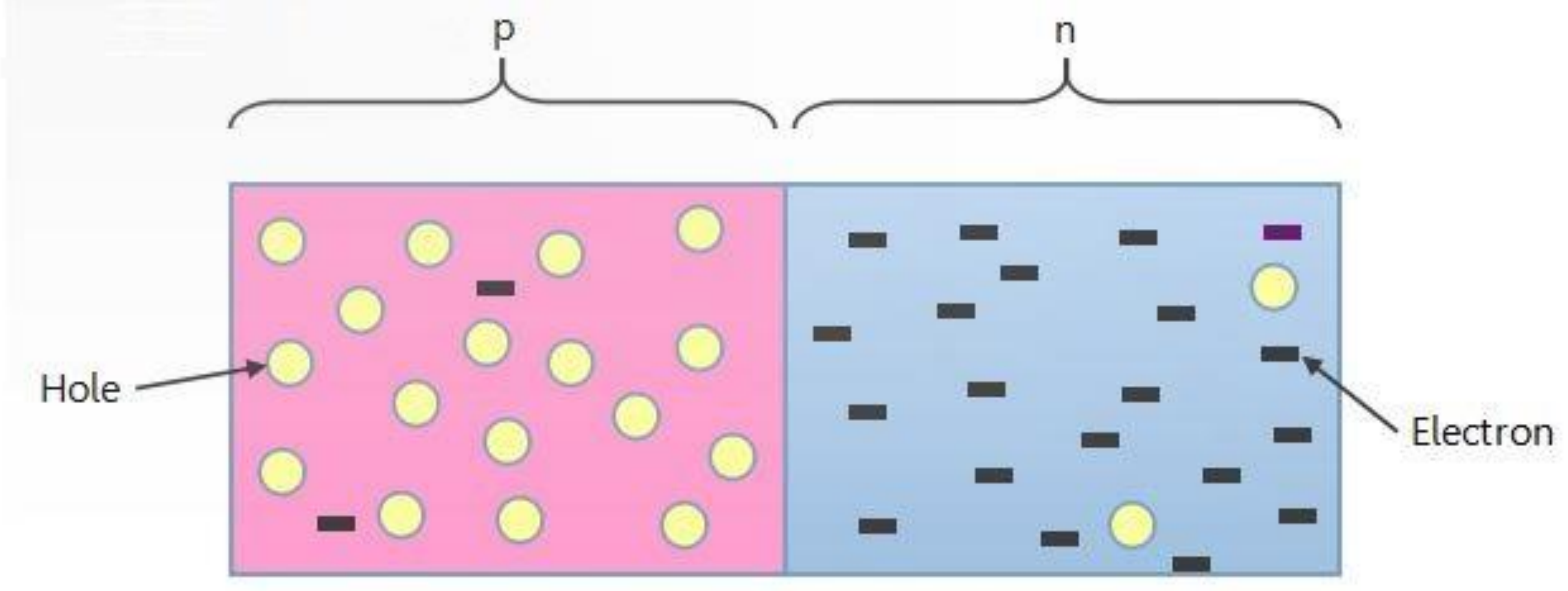
p-n Junction Diode:

The device with a p-n junction is known as a p-n junction diode or in short, a "Diode".



Formation of depletion region:

Diode is formed simply by joining p-type and n-type semiconductor materials.

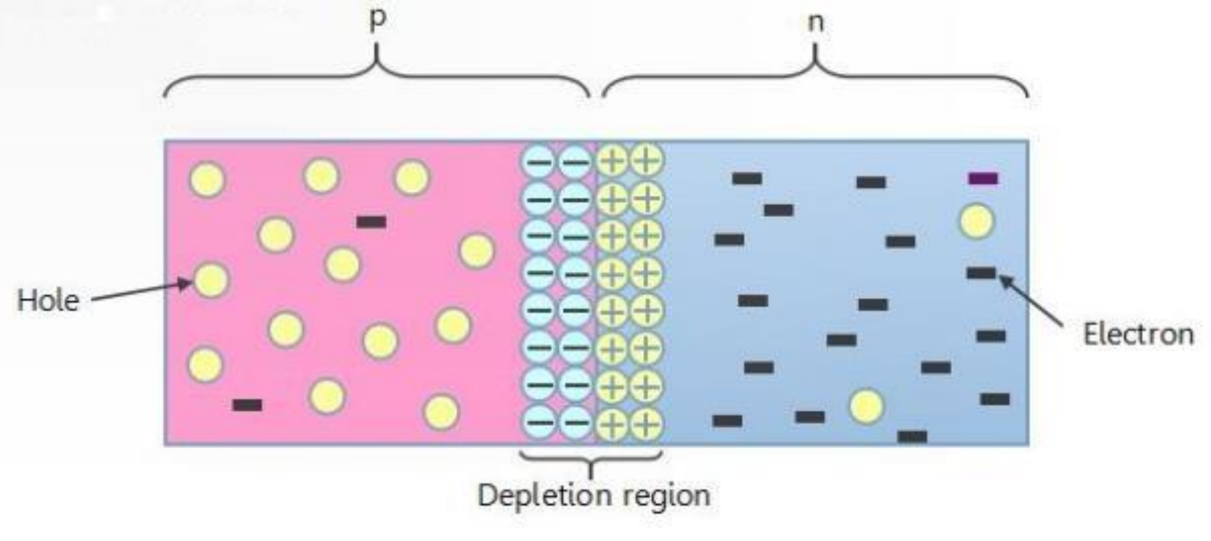


Situation before formation of depletion region

Formation of depletion region:

➢ *The Diffusion Current, I_D:*

Because the concentration of holes is high in the p region and low in the n region, holes diffuse across the junction from the p side to the n side. Similarly, electrons diffuse across the junction from the n side to the p side. These two current components add together to form the diffusion current ID, whose direction is from the p side to the n side.

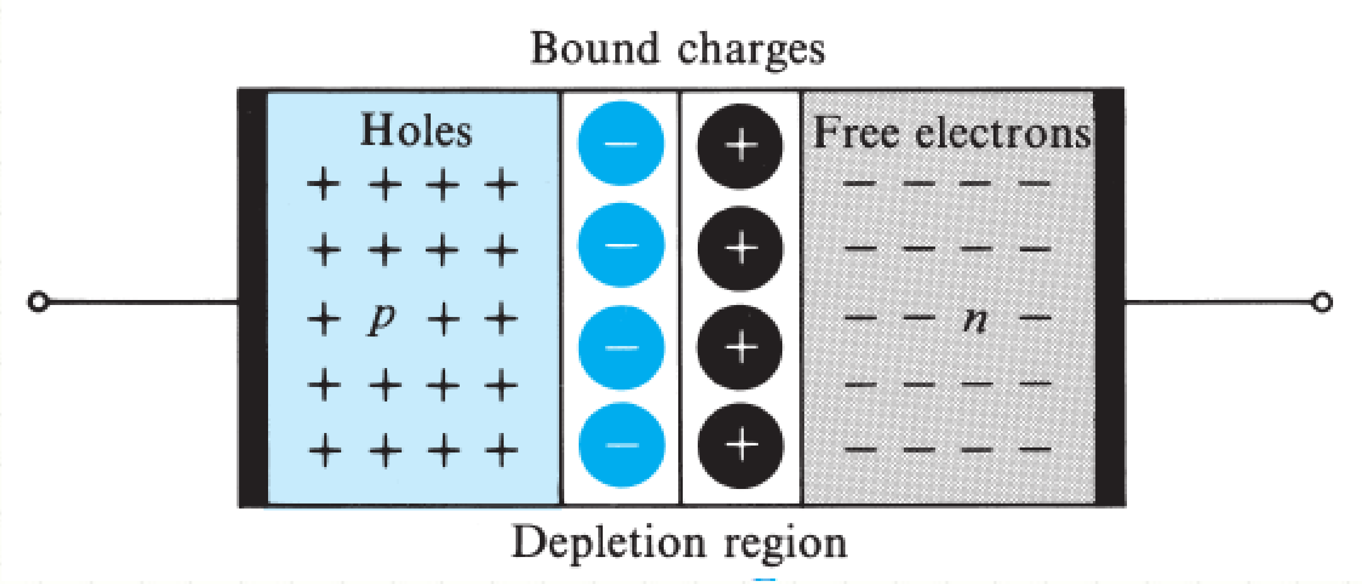


Situation after the formation of depletion region

Formation of depletion region:

➢ ***The Depletion/Space Charge Region:***

The holes that diffuse across the junction into the n region quickly recombine with some of the majority electrons present there and thus disappear from the scene. This recombination process results also in the disappearance of some free electrons from the n-type material.

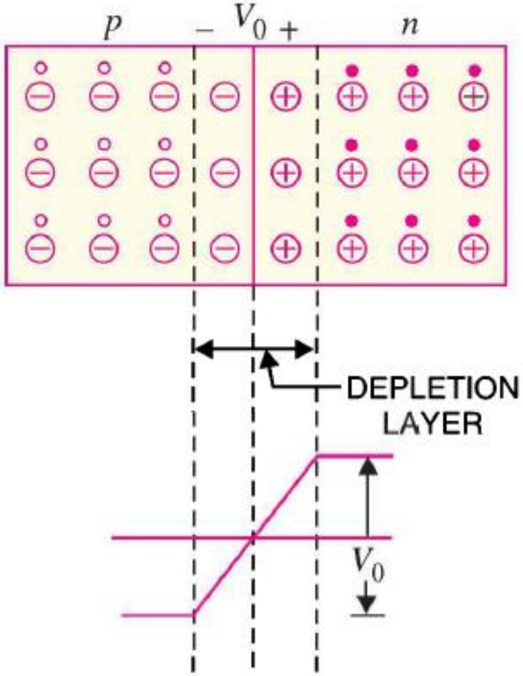


Formation of depletion region:

➢ *The Barrier/Contact Potential:*

The electric field is a barrier to the free electrons in the *n*-region. There exists a potential difference across the depletion layer and is called **barrier potential** (V_0). The barrier potential of a *pn* junction depends upon several factors including the type of semiconductor material, the amount of doping and temperature. The typical barrier potential is approximately:

For silicon, $V_0 = 0.7$ V ; For germanium, $V_0 = 0.3$ V



This built-in potential barrier, or built-in voltage is given by,

$$V_B = \frac{kT}{q_e}\ln\left(\frac{N_a N_d}{n_i^2}\right) = V_T \ln\left(\frac{N_a N_d}{n_i^2}\right)$$

where $V_T \equiv kT/e$, k = Boltzmann's constant, T = absolute temperature, e = the magnitude of the electronic charge, and Na and Nd are the net acceptor and donor concentrations in the P- and N-regions, respectively. The parameter V_T is called the ***thermal voltage*** and is approximately $V_T = 0.026\text{ V}$ at room temperature, $T = 300\text{ K}$.

Thermal Voltage:

Thermal voltage is the voltage generated within a semiconductor junction due to temperature variation..

Thermal voltage is generated in a semiconductor material such as a diode, transistor, MOSFET, etc, due to the thermal energy given to the charge carriers, i.e., electrons and holes. As temperature rises, more charge carriers cross the junction, creating a voltage. This phenomenon is related to the diffusion of charge carriers and is described by the Boltzmann factor, also known as Boltzmann voltage.

Thermal Voltage:

The equation for calculating the thermal voltage V_T is as follows:

$$V_T = \frac{KT}{q}$$

$Where,$

$V_T = Thermal\ Voltage$

$K = Boltzman's\ Constant,$

$(1.3806\ \times 10^{-23}\ eV/k)$

$T = Absolute\ Temperature(Kelvin)$

$q = Charge\ of\ an\ electron$

$(1.602 \times 10^{-19} C)$

Problem:

At a temperature of 27°C (common room temperature), determine the thermal voltage V_T .

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Solution:

$$T = 273 + {}^\circ\mathrm{C} = 273 + 27 = 300\ \mathrm{K}$$

$$V_T = \frac{kT_K}{q} = \frac{(1.38 \times 10^{-23}\ \mathrm{J/K})(30\ \mathrm{K})}{1.6 \times 10^{-19}\ \mathrm{C}}$$

$$= 25.875\ \mathrm{mV} \cong 26\ \mathrm{mV} \quad \text{[Ans.]}$$

Problem:

Calculate the potential barrier of a silicon PN junction, at junction temperature T = 300K. Consider that the doping concentration in P-type region is N_a= 10^{16} cm^{-3} and that in N-type region is N_d= 10^{17} cm^{-3} and the intrinsic carrier concentration in silicon is ni=1.5×10^{10} cm^{-3}.

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Solution:

$$V_B = V_T \ln\left(\frac{N_a N_d}{n_i^2}\right) = 26 \times 10^{-3}\ \mathrm{V} \ln\left[\frac{(10^{16}\ \mathrm{cm}^{-3})(10^{17}\ \mathrm{cm}^{-3})}{(1.5 \times 10^{10}\ \mathrm{cm}^{-3})^2}\right]$$

$$= 26 \times 10^{-3}\ \mathrm{V} \ln\left[\frac{10^{13}}{1.5 \times 1.5}\right] \approx 757.12 \times 10^{-3}\ \mathrm{V} \models 0.757\ \mathrm{V}\ [\mathrm{Ans.}]$$

Biasing of p-n Junction

Biasing of p-n Junction:

There are two operating regions and three possible “biasing” conditions for the standard **Junction Diode** and these are:

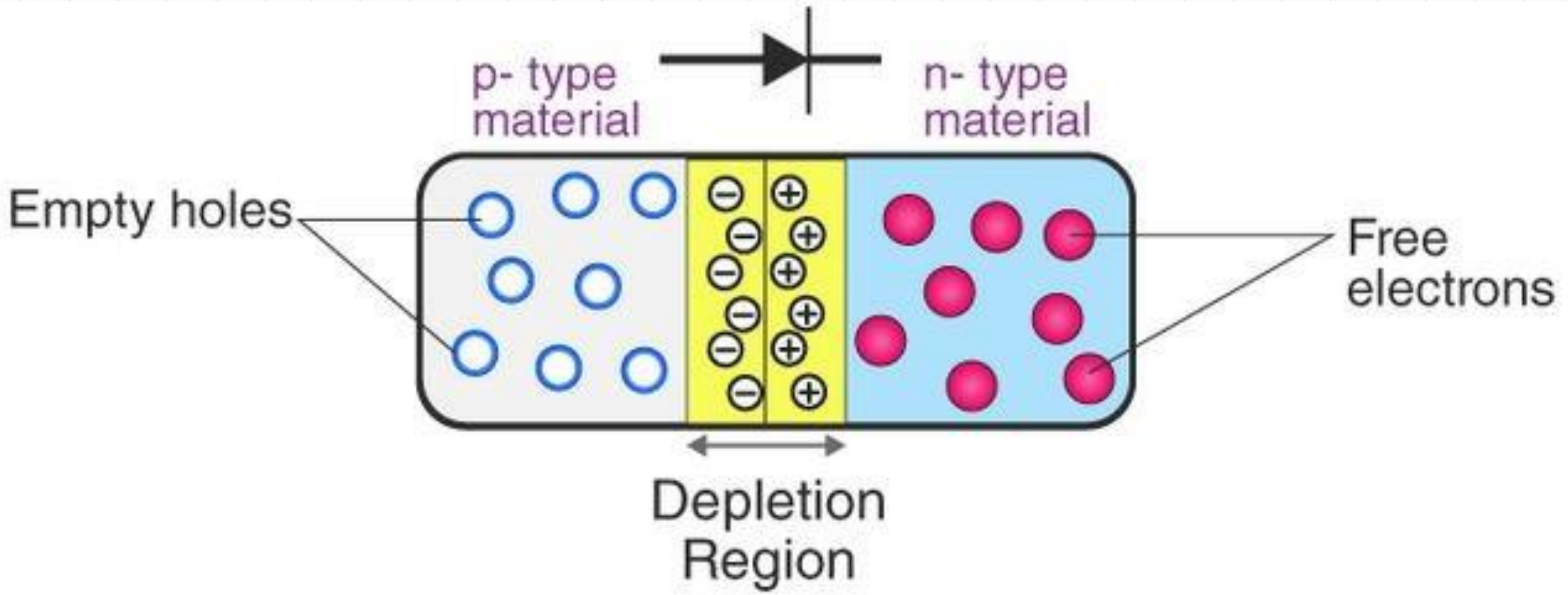
1. Zero Bias: If no external voltage is applied to the p-n junction diode, it is said to be in zero bias condition.

2. Forward Bias: If the voltage is connected as negative (-ve) to the n-type material and positive (+ve) to the p-type material across the diode, then the diode is said to be in forward bias condition.

3. Reverse Bias: If the voltage is connected as positive (+ve) to the n-type material and negative (-ve) to the p-type material across the diode, then the diode is said to be in reverse bias condition.

Zero bias:

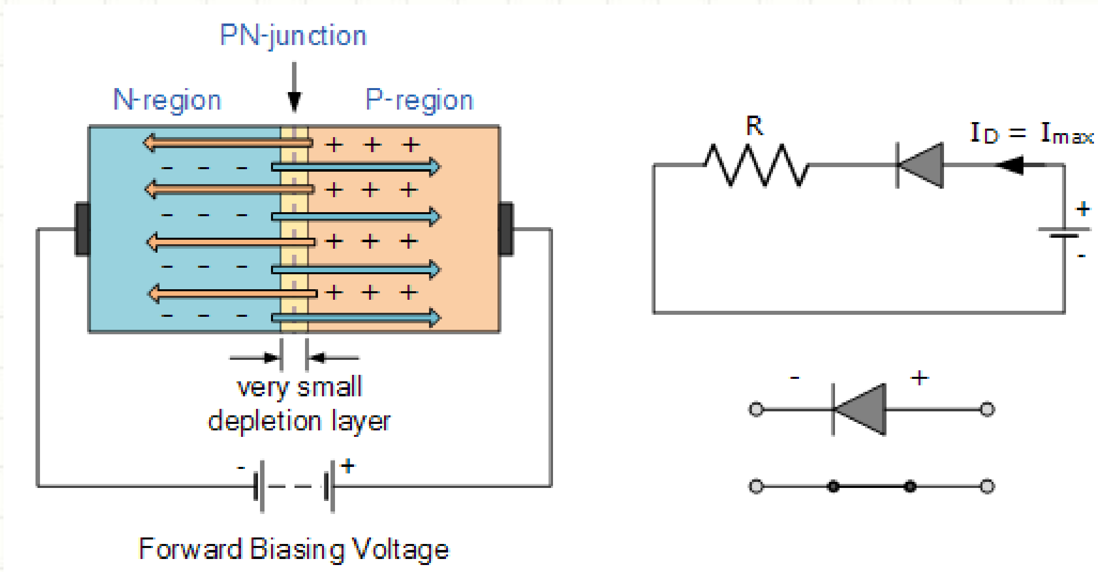
When a diode is connected in a Zero Bias condition, no external potential energy is applied to the PN junction. Once p-n junction is formed and depletion layer is created, the diffusion of electrons and holes stops. In other words, the depletion layer acts as a barrier to the further movement of free electrons across the junction.



Forward bias:

When the p-type is connected to the battery's positive terminal and the n-type to the negative terminal, then the P-N junction is said to be Forward biased. When the P-N junction is forward biased, the built-in electric field at the P-N junction and the applied electric field are in opposite directions. The applied electric field has a greater magnitude than the built-in electric field. This results in a less resistive and thinner depletion region.

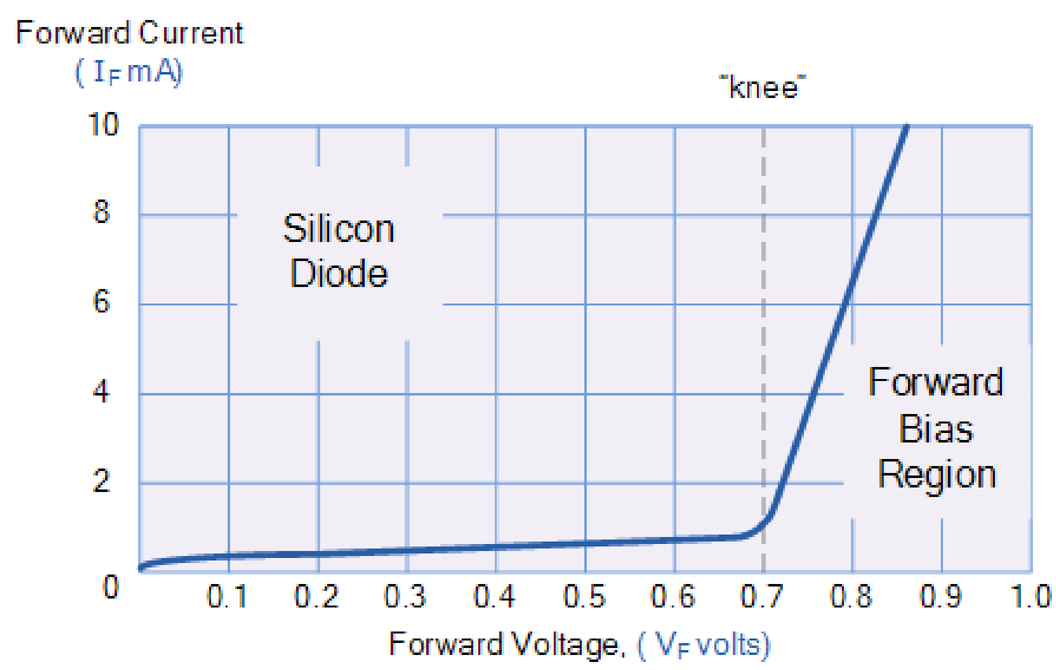
In silicon, at the voltage of 0.7V; and in Germanium, at the voltage of 0.3V, the resistance of the depletion region becomes completely negligible and the current flows across it rapidly and unimpeded.



Forward Characteristics Curve:

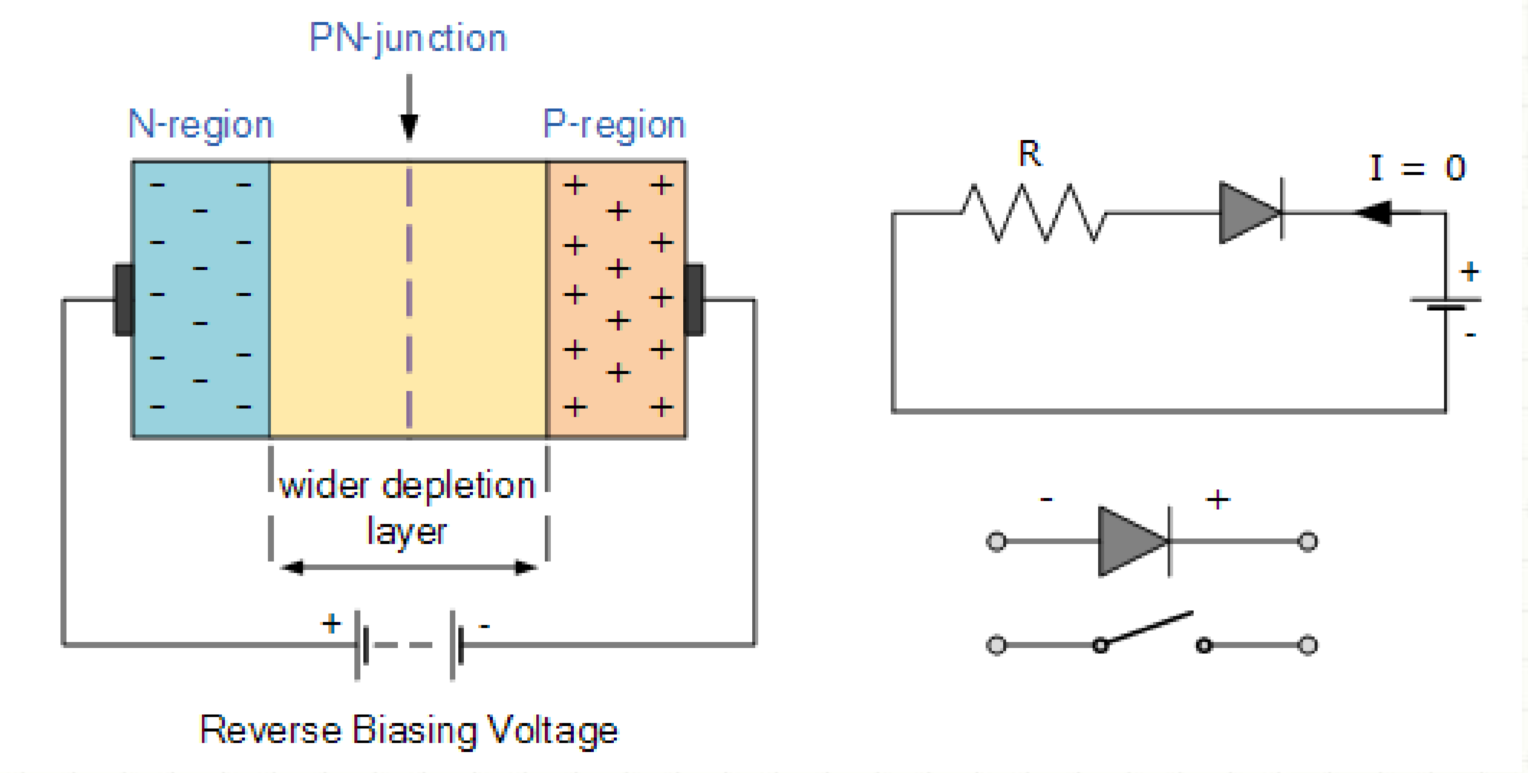
When a diode is connected in a Forward Bias condition, if the external voltage becomes greater than the value of the barrier potential (approx. 0.7 volts for silicon and 0.3 volts for germanium), the potential barriers opposition will be overcome and current will start to flow.

This results in a characteristics curve of a very little current flowing up to this voltage point, called the **"knee"** on the static curves and then a high current flow through the diode with little increase in the external voltage.



Reverse bias:

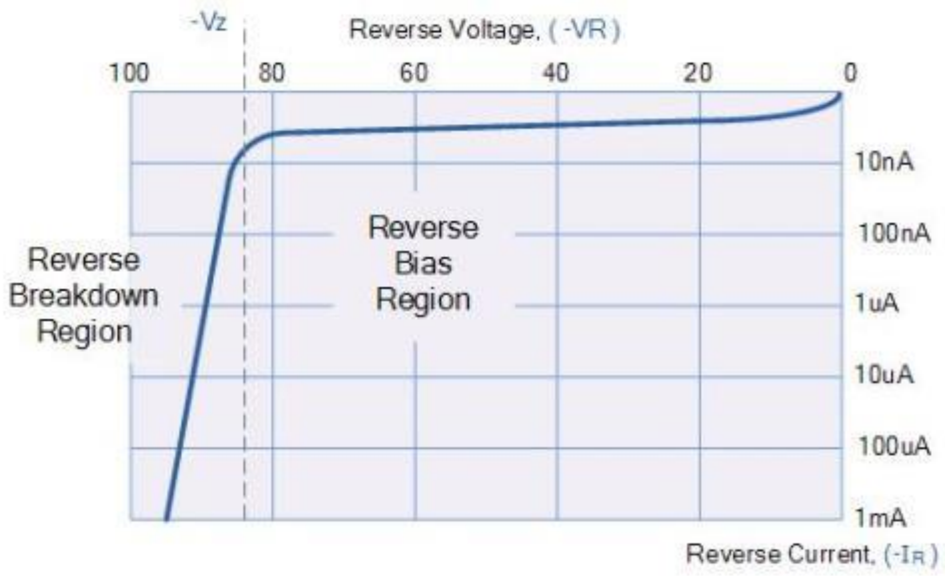
When the p-type is connected to the battery's negative terminal and the n-type is connected to the positive side, the p-n junction is reverse biased. In this case, the built-in electric field and the applied electric field are in the same direction. When the two fields are added, the resultant electric field is in the same direction as the built-in electric field, creating a more resistive, thicker depletion region.



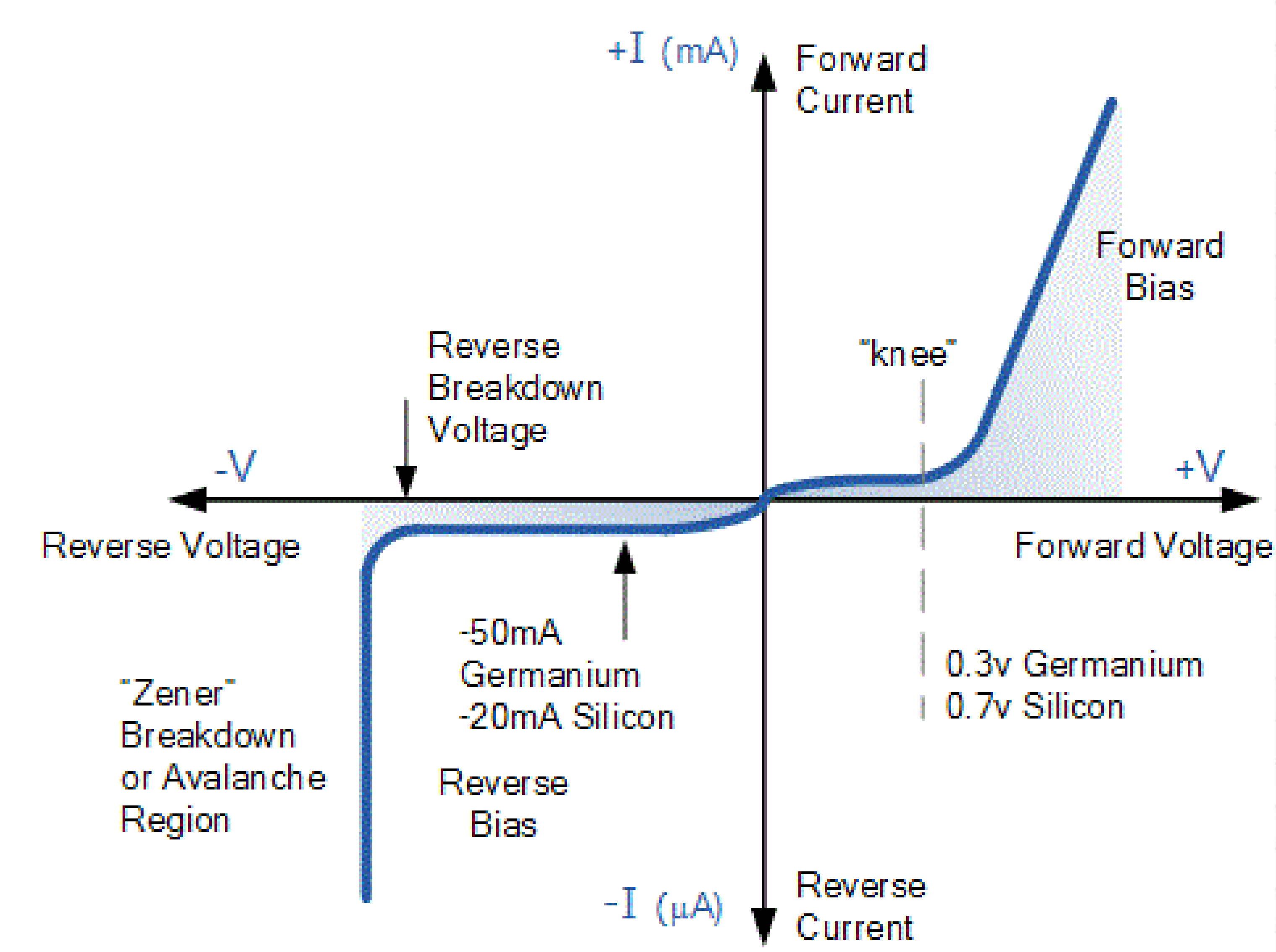
Reverse Characteristics Curve:

As the depletion layer grows wider due to a lack of electrons and holes and presents a high resistive path, a high potential barrier is created across the junction. thus practically zero current flows through the junction diode with an increase in bias voltage. However, a very small reverse leakage current does flow through the junction which can normally be measured in micro-amperes (μA).

One final point, if the reverse bias voltage, Vr applied to the diode is increased to a sufficiently high enough value, it will cause the diode's PN junction to overheat and fail due to the avalanche effect around the junction. This may cause the diode to become shorted and will result in the flow of maximum circuit current.



I-V Characteristics Curve:



I-V Characteristics Curve:

Knee Voltage:

It is the forward voltage at which the current through the junction starts to increase rapidly.

Breakdown Voltage:

It is the minimum reverse voltage at which p-n junction breaks down with sudden rise in reverse current.

Avalanche Breakdown:

The reverse bias increases the electrical field across the depletion region. When the high electric field exists across the depletion, the velocity of minority charge carrier crossing the depletion region increases. The process is continuous and the electric field becomes so much higher then the reverse current starts flowing in the PN junction. The process is known as the **Avalanche breakdown**. After the breakdown, the junction cannot regain its original position because the diode is completely burnt off.

I-V Characteristics Curve:

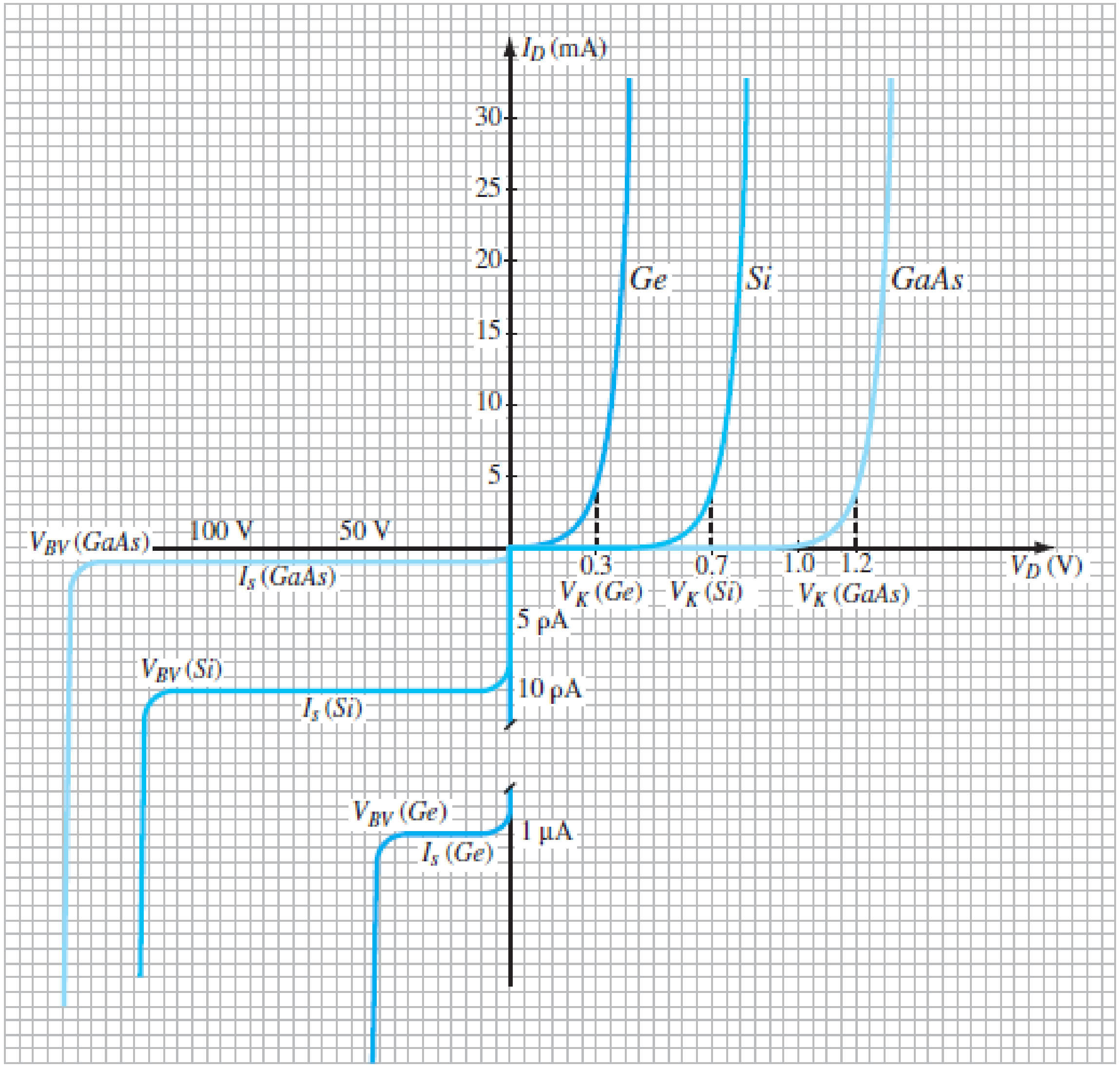


FIG. 1.18
Comparison of Ge, Si, and GaAs commercial diodes.

QUESTIONS?

See You In The Next Class